

III. Basic Properties of polymers

3.1 Introduction

Intermolecular force :

Molecular interaction in polymer affect to chain orientation and many properties of polymer.

Molecular interaction in polymer are:

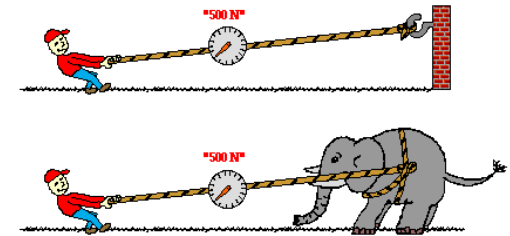
(i) London dispersion force

Weak intermolecular interaction that presence in **non-polar molecules**.

This force is quiet important in hydrocarbon polymer such as PP, PE, PS

Even, it's weak interaction but along the long chain, this force can hold chain together.

Key point : the chain need to be close pack.



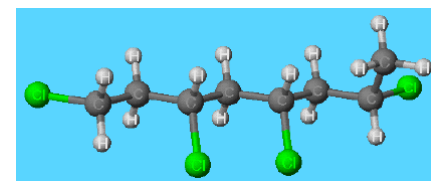
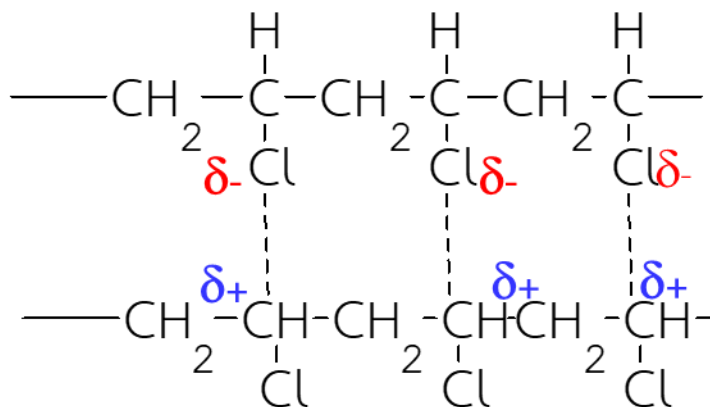
(ii) Dipole force

Attractive forces between the positive end of one polar molecule and the negative end of another polar molecule.

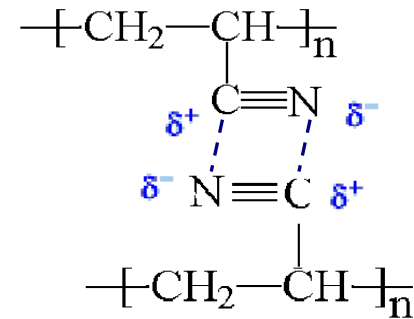
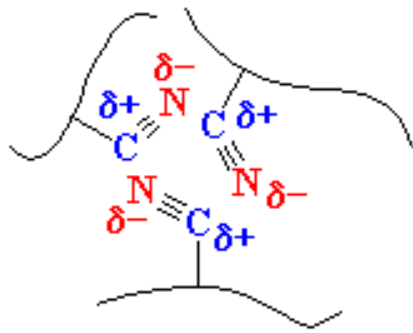
It have a significant effect only when the molecules involved are close together.

Occur in polar polymers.

e.g. poly(vinyl chloride)

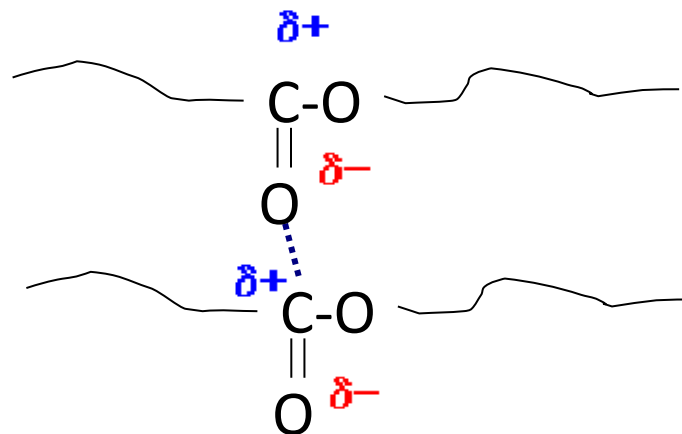


Polyacrylonitrile



Polyester

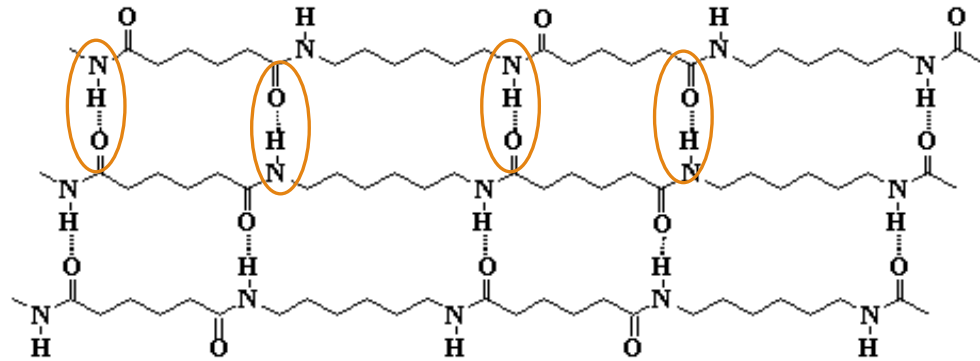
Interaction of C and O in C=O group



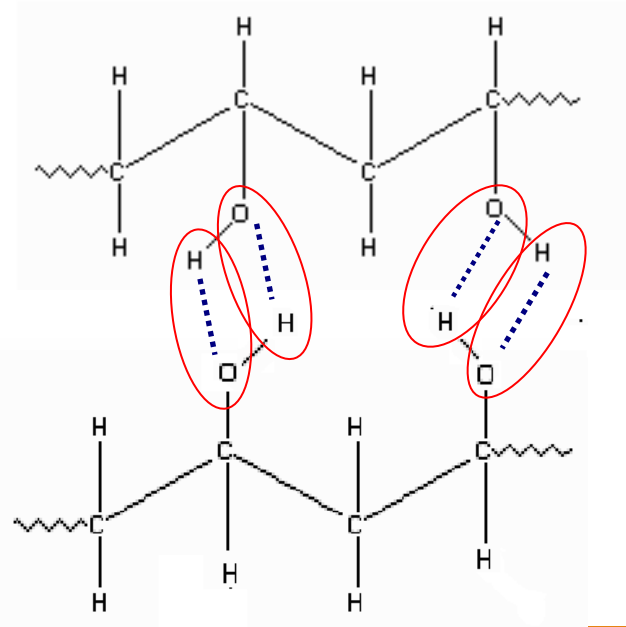
(iii) Hydrogen bond

Attractive force of N or O and H atom in polymer molecules.

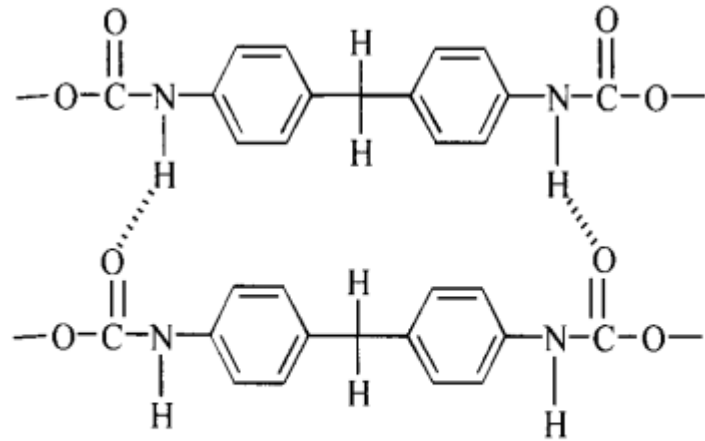
e.g. polyamide



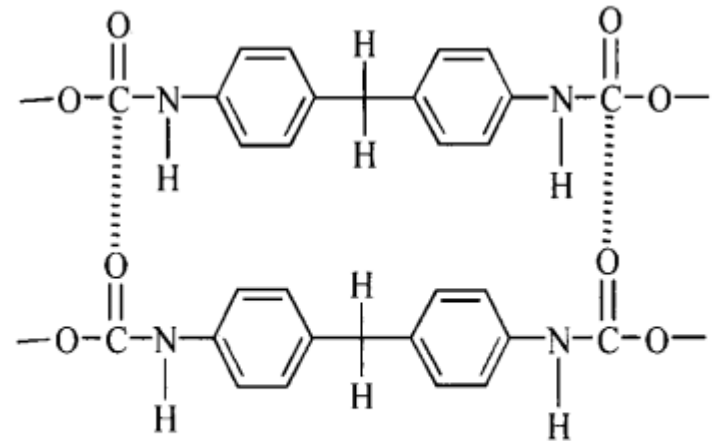
Poly(vinyl alcohol)



Polyurethane



(1) hydrogen bonding

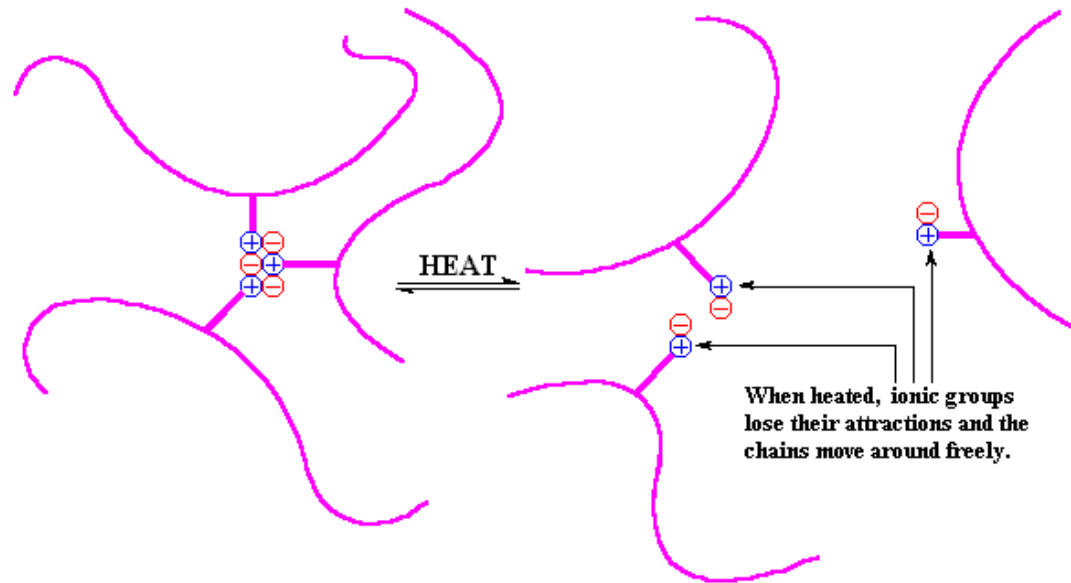


(2) dipole-dipole interaction

(iv) Coulombic force, in Ionomers

Electronic charge interaction (ionic force)

Ionomers : polymer comprising of repeat units contain ionic pendent group (<15 %).

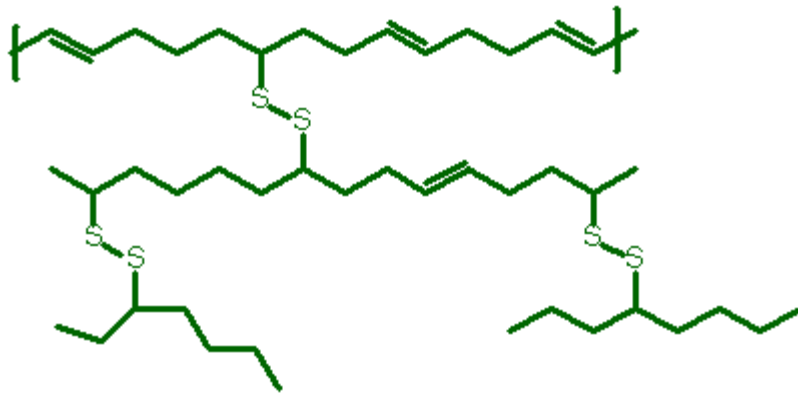
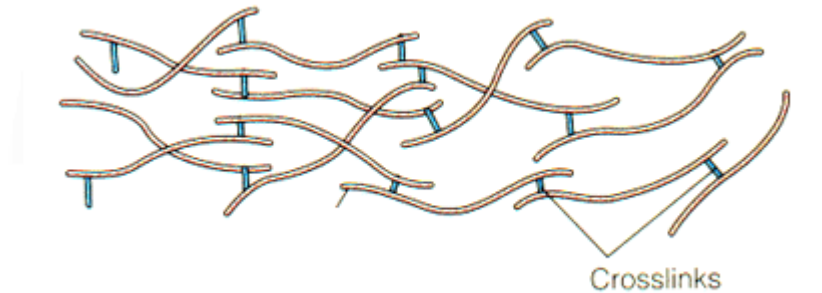


Reversible physical crosslink

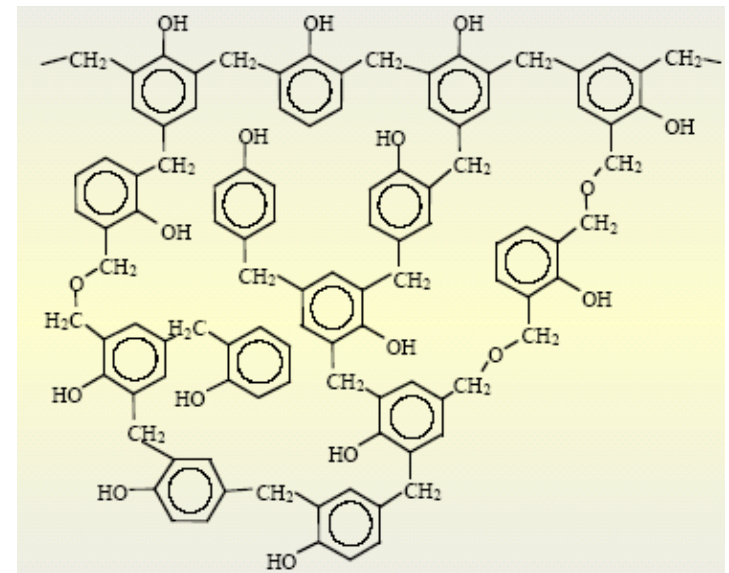
(V) Crosslink

Covalent bond (**not exactly interaction**) that link one polymer chain to another.

Thermosetting p



Vulcanized rubber



Phenolic resin

Strength Of interaction

- 
- Hydrocarbons -- PE, IPP, PS, etc.
 - Weak Dispersion Forces
 - Polar Polymers -- PVC, PAN, etc.
 - E.g.: Those Containing Heteroatoms (O, N, Cl, F)
 - Dipole/dipole Interactions
 - Hydrogen Bonding Polymers
 - E.g.: Nylons, Polyurethanes, etc.
 - Ionomers
 - Crosslink

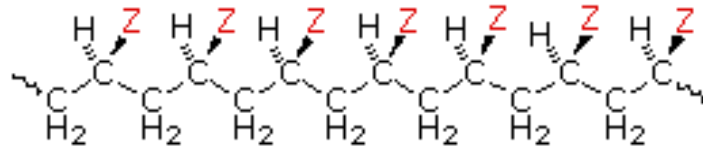
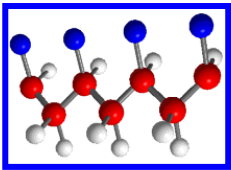
Increasing
Interaction
Strength

Linear Vinyl Polymer with head-to-tail structure:

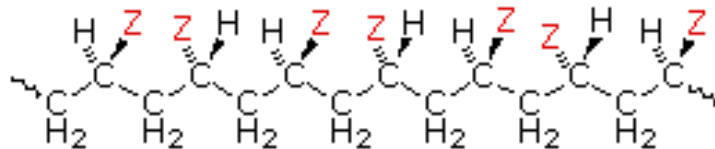
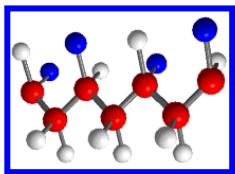
3 types of z arrangement

หมู่ Z สามารถต่อกับอะตอมของคาร์บอนได้ 3 แบบ เกิดเป็น isomer ที่เรียกว่า "stereoisomer" (tacticity) ทำให้ได้โมเลกุลที่มี configuration แตกต่างกันดังนี้

1. isotactic (ไอโซแทกติก) : หมู่แทนที่อยู่ด้านเดียวกัน



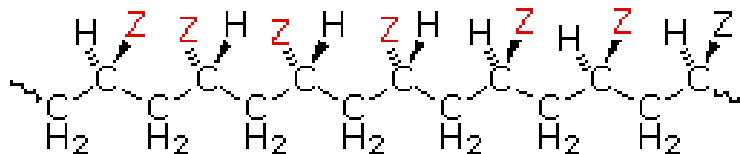
2. syndiotactic (ซินดิโอแทกติก) : หมู่แทนที่อยู่สลับกัน



isotactic & syndiotactic

↓
เป็นโครงสร้างที่สม่ำเสมอ และ
มีระเบียบ เรียกว่า
"stereoregularity" จึงมีแนวโน้มที่
จะเกิดผลึกได้ (crystalline polymer)

3. atactic (อแทกติก) : หมู่แทนที่อยู่สลับกันแบบไร้ระเบียบ



← amorphous polymer

3.2 Solubility of Polymer



Polymers are giant molecule (long chain structure), hence their solubility are different from small molecules,

i.e. slowly dissolve and the solvents are limited

“ Like dissolves like ”

Polar polymers : poly(acrylic acid), polyacrylamide,
poly(vinyl alcohol)



soluble in water

Nonpolar polymers/low polarity : polystyrene,
poly(methyl methacrylate), poly(vinyl chloride)

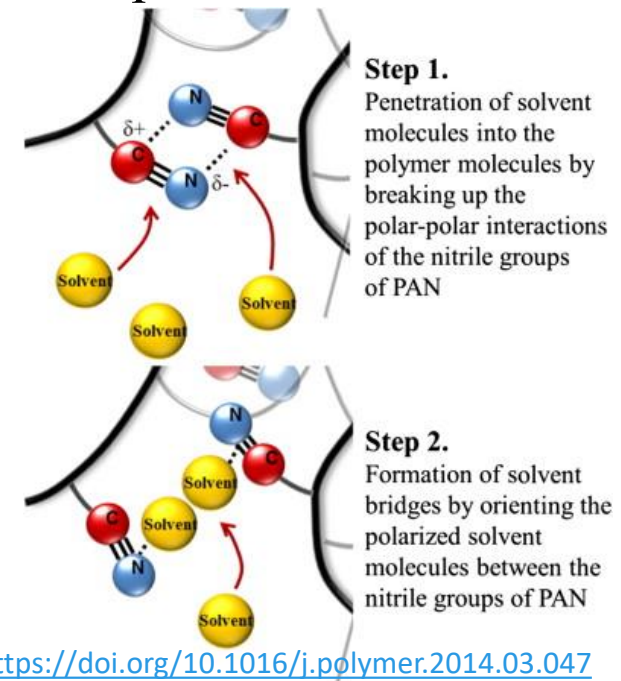


soluble in nonpolar solvents

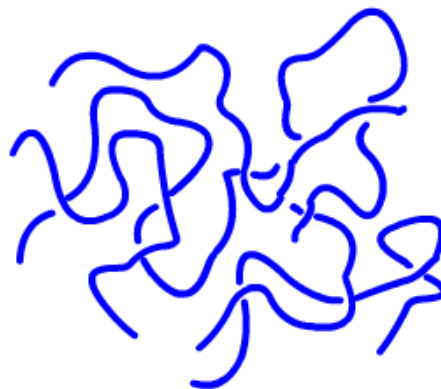
The solubility of the polymer in a good solvent has 2 steps.

1. **Solvent** molecules **penetrate** between polymer molecules ➡ **swollen polymer (like gel)**, but still heterogeneous.

2. **Gel structure is destroyed**, and polymer molecules dispersion in solvent until homogenous solution is obtained. **Polymer molecule interact with solvent molecules.**



a) Polymer molecules in solid state just after being added to a solvent



b) First step: a swollen gel in solvent

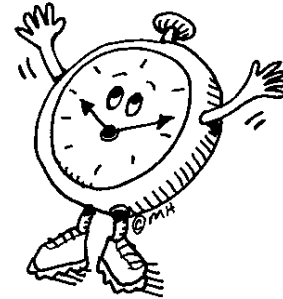


c) Second step: solvated polymer molecules dispersed into a solution

Polymer may soluble in one solvent, but insoluble in other solvents.

Solubility can be improved by:

- Give more time
- increase temp.

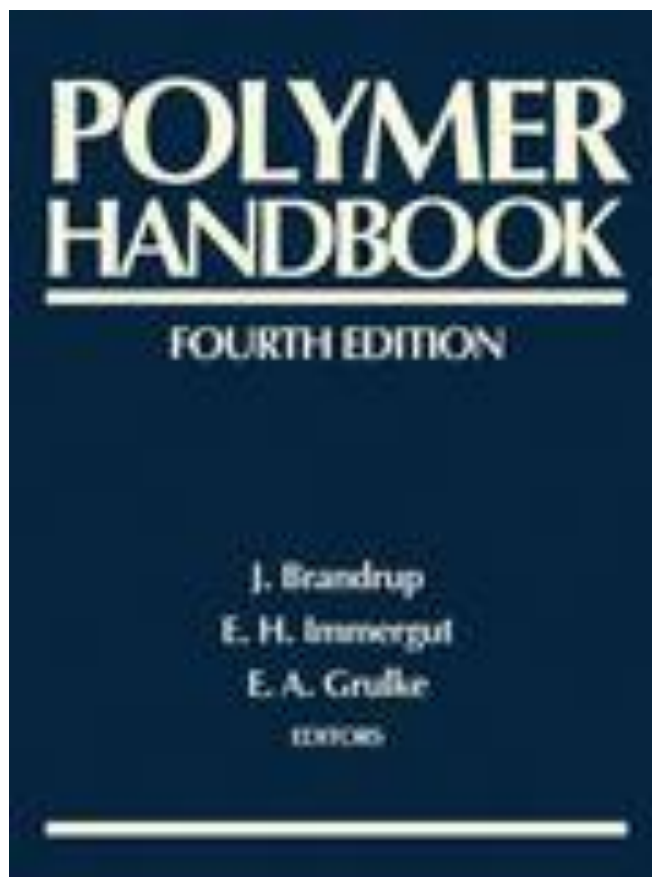


In the same kind polymer, **branched structure** have higher solubility than **linear structure**.

Cross-linked polymer is **insoluble** in any solvent, it can **only be swell**.

Degree of swelling depends on crosslink density of those polymers.

Guideline to choose a suitable for polymers “**Polymer Handbook**”



Polymer Handbook; Eds. Brandrup, J.; Immergut, E.H.; Grulke, E.A., 4th Edition, John Wiley, New York, 1999

Solubility parameter of polymers and solvents

Polymer	δ (cal/cm ³) ^{1/2}	Solvent	δ (cal/cm ³) ^{1/2}
Polyacrylonitrile	12.5	Acetone	9.9
Polyethylene	8.0	Benzene	9.2
Polystyrene	8.7	Carbon tetrachloride	8.6
Poly(methyl methacrylate)	9.5	Chloroform	9.3
Polypropylene	9.3	Methanol	14.5
Poly(vinyl alcohol)	12.6	Tetrahydrofuran	9.1
Poly(vinyl chloride)	9.5	Toluene	8.9
		Water	23.4

Polymer is able to dissolve in solvent at critical value

e.g. polystyrene are soluble in many solvents

$$(\delta_1 - \delta_2)^2 < 3$$

except water and methanol

Remark : This criteria use for **non-polar & amorphous polymer** and polymer **without H-bonding interaction**

Ex. Polyethylene and polypropylene should be soluble in non-polar solvent, both polymers are insoluble because of crystallinity. It become soluble at high temp. (near its T_m)

PE ($T_m \sim 130^\circ\text{C}$) และ PP ($T_m \sim 160^\circ\text{C}$)

Dissolution of polymer is very slow, it can take for a few day.

Factors affected to solubility are:

- ✓ **Polarity** : polar polymer trend to dissolve in polar solvent
- ✓ **molecular weight** : solubility decrease as MW increase
- ✓ **Branching**: branched structure polymer is easily dissolve than linear one
- ✓ **Crosslinking** : x-link polymer is not dissolve in any solvent, it only swell. High degree of x-link, low swelling ratio.
- ✓ **Crystallinity** : high crystallinity polymer is more difficult to dissolve, lower solubility

3.3 Solid state of polymer

Polymer classification following morphology :

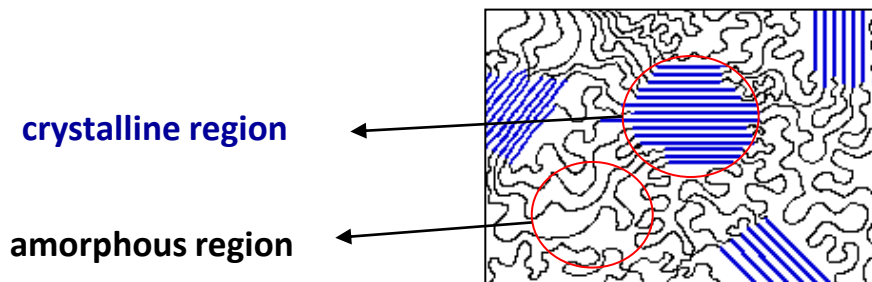
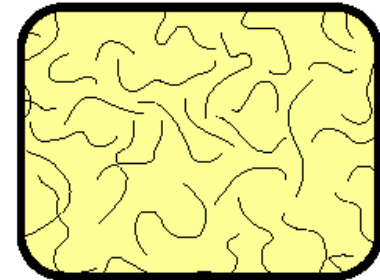
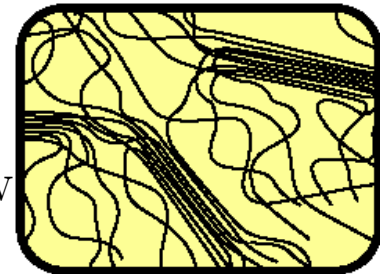
- Crystalline polymer or semicrystalline :

Some part of polymer chain are well oriented and interacted with others.

- Amorphous polymer :

Polymer chains are unoriented.

There no polymers is complete crystalline.



Polymer $\Leftarrow$ crystalline region
+
amorphous region

Degree of crystallinity (or % crystallinity)

< 20% : low crystalline polymer

20–60% : medium crystalline polymer

>60 % : high crystalline polymer

Degree of Crystallinity depends on:

➤ Intermolecular force

High Interaction, good chain attachment and high crystallinity.

➤ Polymer structure

Chemical structure results in the interaction of polymers and crystallinity.

Linear polymer is higher crystallinity than branched polymers.

➤ Stereoregularity/tacticity

Structural regularity results in easily packing of molecular chain, hence high regularity structure could be high crystallinity polymers, such as :

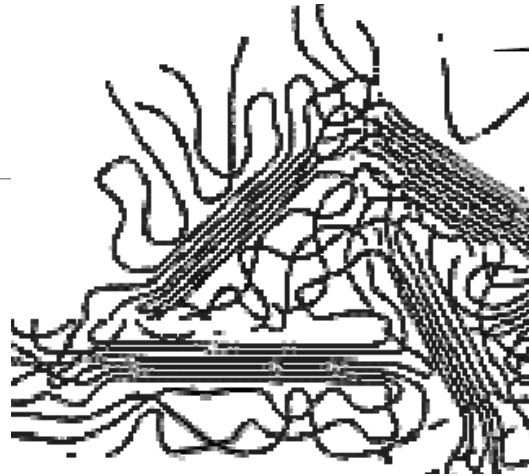
- Isotactic and syndiotactic polymer have a higher crystalline than atactic.
- Homopolymer is higher crystallinity than copolymers
- Alternating and block copolymer is higher crystallinity than random copolymers.

➤ Mol. Wt.

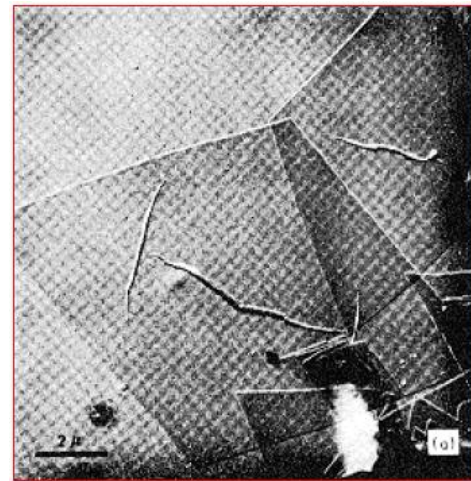
High MW, low crystallinity

Crystalline polymer model

Fringed Micelles

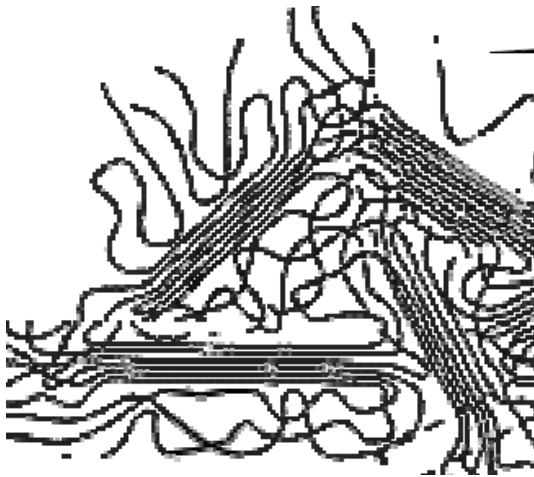


Single Crystal Lamellae



Fringed Micelles

: melt crystallization (solid state crystalline)



The 1st Really Useful Model

Crystalline region disperse in amorphous region

Used to explain a difference between amorphous and crystalline polymers properties

- @ below T_g , polymers are hard and brittle.
- @ above T_g , amorphous polymers is soft and weak, while crystalline polymers is still strong and tough. This because it consist of crystalline (reinforce) part.

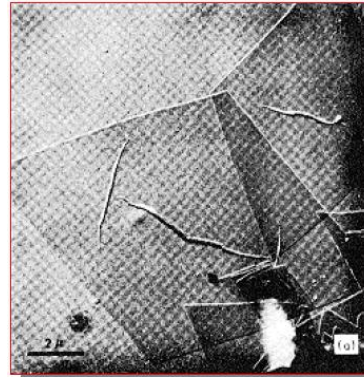
e.g. PE and isotactic PP fiber is strong even @ $T > T_g$, while actactic PP is gum.

« เอกสารฉบับนี้ใช้เพื่อการเรียนการสอนเท่านั้น »

Single Crystal Lamellae

Dilute solution crystallization

Electron Micrograph
of a Polyethylene
Solution Grown
Single Crystal

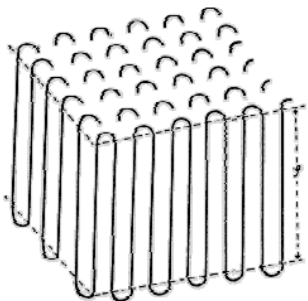


Electron diffraction analysis : thickness of crystal laminate is 100–200 °A

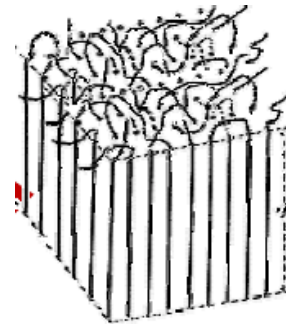
How polymer chain (2000 °A length) is inside?

It proposed that polymer chain are folded perpendicularly with crystal planar.

Chain folded model in single crystal



Regular Folding Chain
(Adjacent Re-entry)

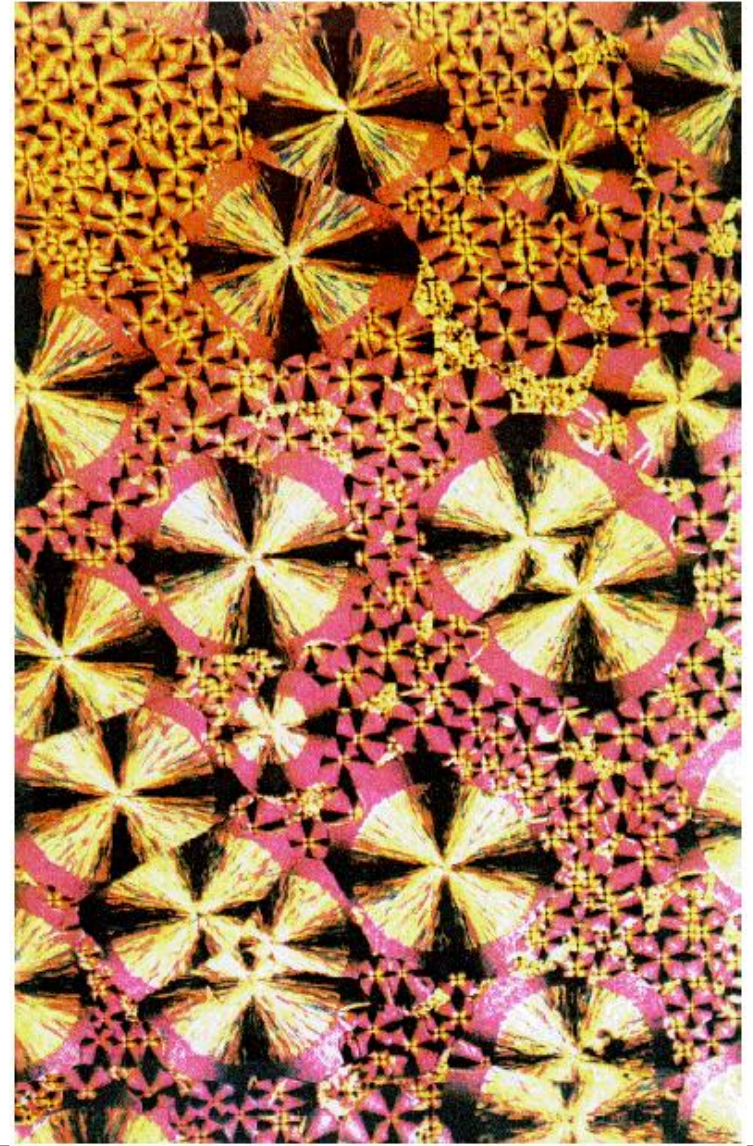
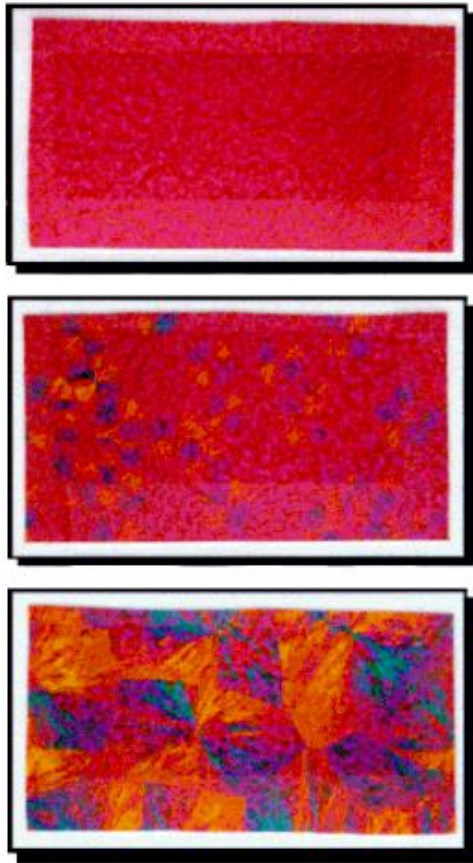


Irregular Chain Folding
(Random Re-entry)
“ switchbord model”

SPHERULITES

are spherical semicrystalline regions inside polymers. Their formation is associated with crystallization of polymers from the melt.

Spherulites can see under optical microscope (OM).



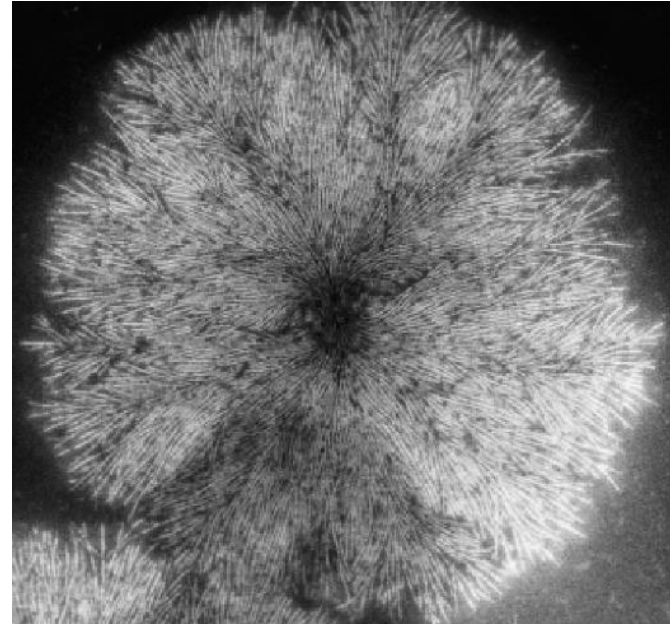
SPHERULITES

: Grow up from center and spread out in radius direction

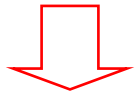
start (nucleate/initiate)



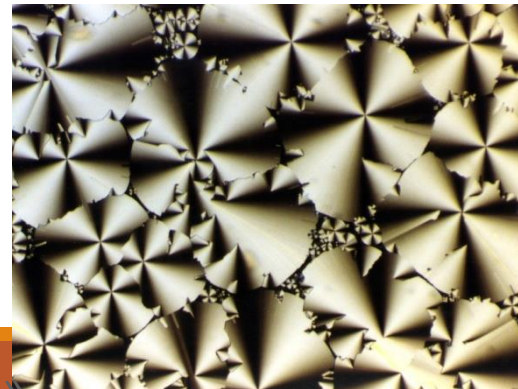
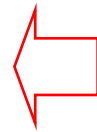
grow (spherically)



overlap (impinge)



grow more
to cover all space

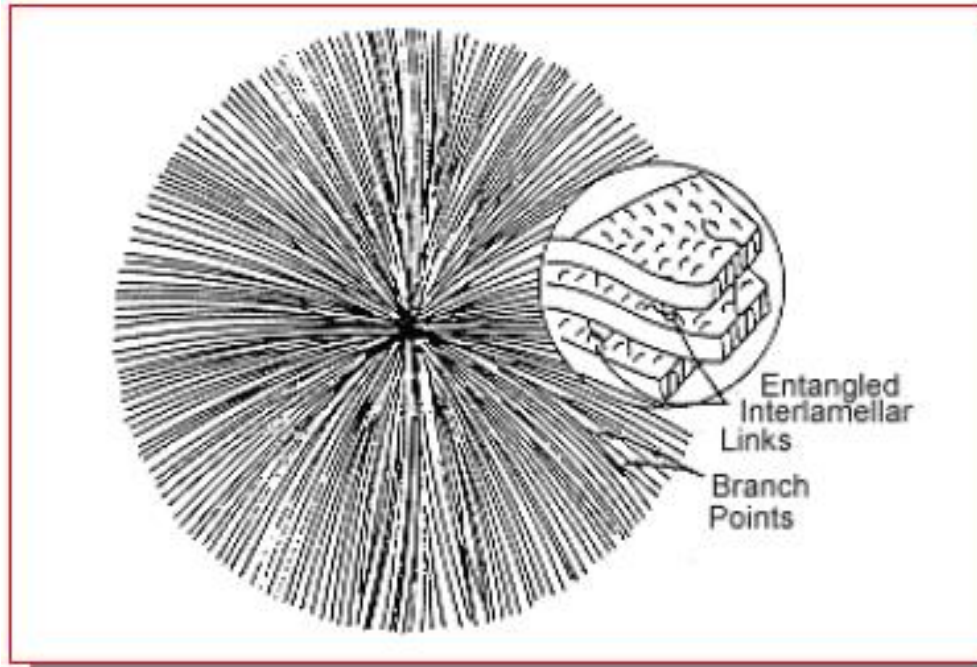


<http://www.youtube.com/watch?v=130sUnjUxmQ>

Structure of Spherulites

Spherulites are composed of highly ordered lamellae, which result in higher density, hardness, but also brittleness of the spherulites as compared to disordered polymer.

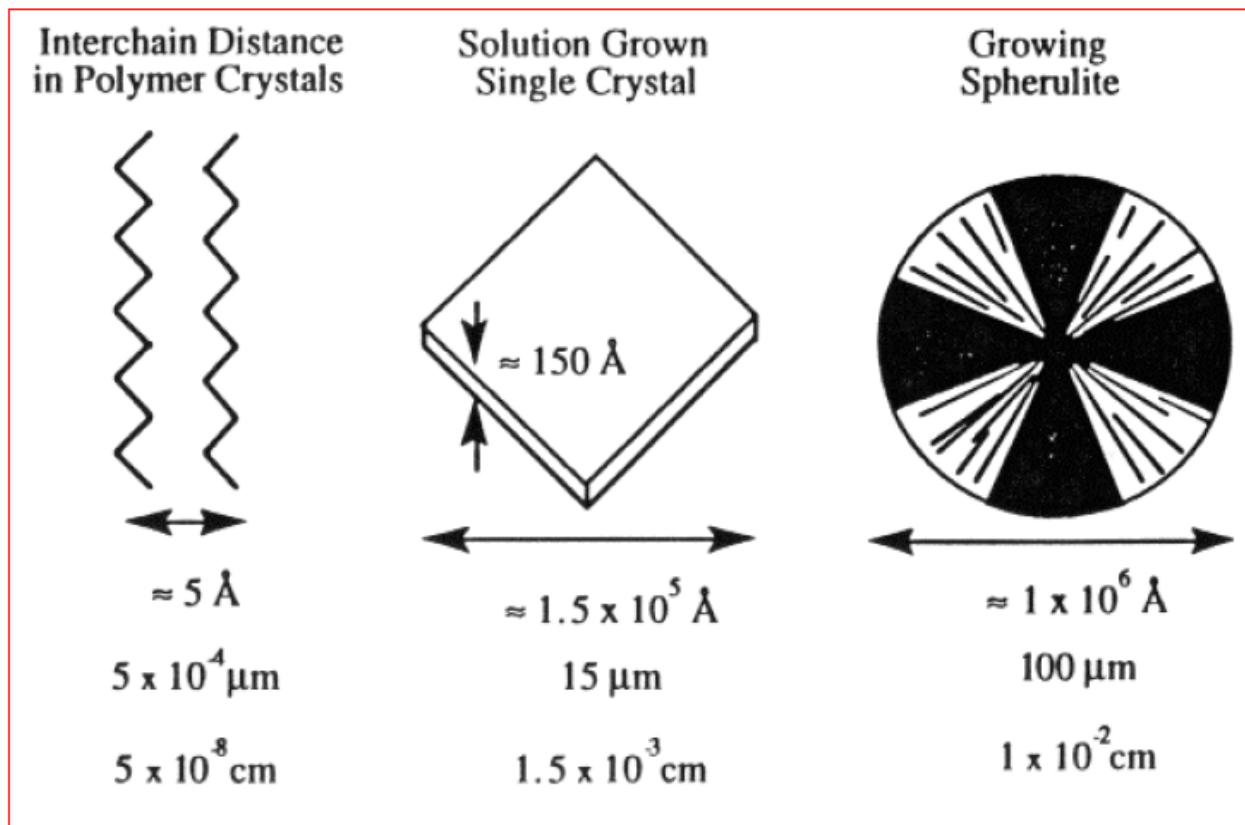
The lamellae are connected by amorphous regions which provide certain elasticity and impact resistance



A Model for the Structure of Spherulites

Lamellae & Spherulites

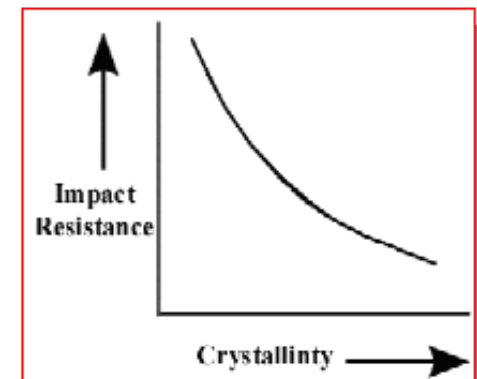
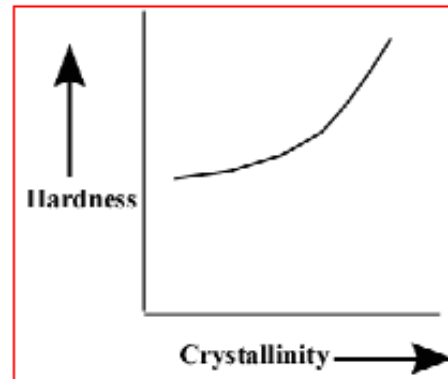
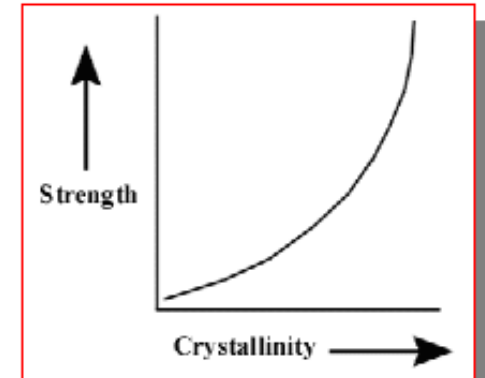
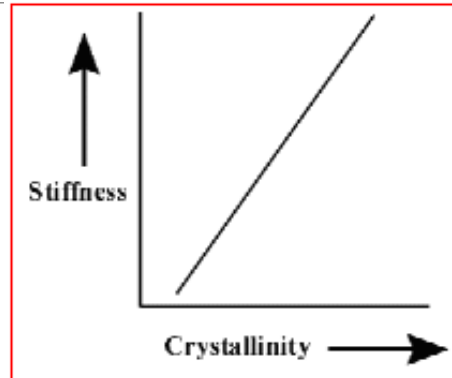
How Big Are They?



Properties of Polymers

Crystallization: Its Relationship to Mechanical Properties

Increasing the Degree of Crystallinity Produces a Stiffer, Harder, Stronger Material. But, the Impact Resistance Decreases

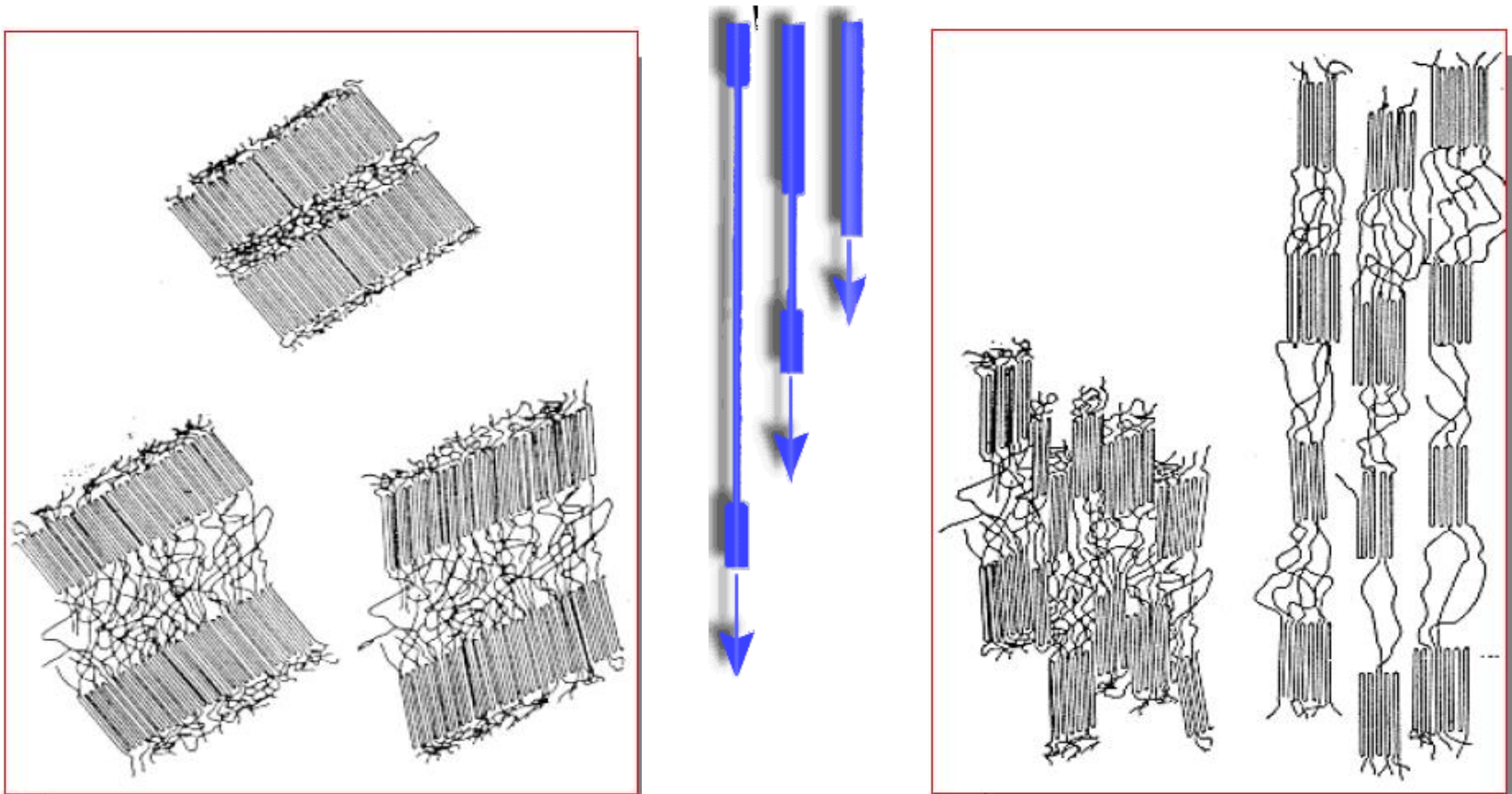


e.g.

the differences in the physical properties of a polyethylene bucket (relatively high crystallinity) and a garbage bag (relatively low crystallinity).

Force -induced crystallization: Fibers

In fiber production, crystalline polymer are drawn out in order to enhance crystallinity. During drawing out, crystalline and amorphous part are reoriented along the force. It make polymer are more crystallinity.



Amorphous polymer

The **molecular entanglements** network in glassy state is inherited from the melt with mobility of the chain drastically reduced. It make some amorphous polymer are certainly strong.

